

Practice Quiz 02D — MATH 2055 — Fall 2026 — Solutions

Practice for Quiz 02

Free variables, bound variables, and renaming.

1. A student claims that renaming the bound variable x as y changes

$$\forall x \in \mathbf{R} (x = y) \quad \text{into} \quad \forall y \in \mathbf{R} (y = y)$$

without changing the meaning.

- (a) Identify the free and bound variables in each formula, and explain what is wrong with the student's claim.

Solution In the first formula, x is bound and y is free. In the second, y is bound and there are no free variables. The proposed replacement binds the previously free occurrences of y , so it changes the meaning: the first formula says every real number equals the given y , whereas the second says each real number equals itself. ■

- (b) Correctly rename the bound variable of the first formula as t .

Solution The correct result is $\forall t \in \mathbf{R} (t = y)$. The free variable y remains free, and the new bound variable t takes the place of x . ■